Online Mean and Variance Computations by Kalman Filtering

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Setup

Suppose that we receive a stream of data $\{x_t : t \geq 1\}$, and we want to compute the sample mean and sample variance online in a recursive and stable manner. One can do this using the Kalman Filter.

The construction is as follows: posit a nominal prior distribution $\mu \sim \mathcal{N}\left(m_0, s_0^2\right)$, a likelihood $x_t \overset{\text{iid}}{\sim} \mathcal{N}\left(\mu, \sigma^2\right)$, and define a scaling factor $\theta \geq 0$ so that $s_0^2 = \theta^{-1} \cdot \sigma^2$. (We will eventually be able to do away with treating any of these objects as 'serious' from a modelling perspective, but they aid in the motivation and derivations.)

Routine computations then yield that the filtering distribution for μ at time t takes the form (writing $\mathscr{F}_t$ for the sigma-algebra generated by $\{x_s : 1 \leq s \leq t\}$)

$$\begin{aligned} \mathbf{P}\left(\mathrm{d}\mu \mid \mathscr{F}_t\right) &= \mathcal{N}\left(m_t, s_t^2\right) \\ \text{where} \qquad \left(m_t, s_t^2\right) &= \left(\frac{t \cdot \bar{x}_t + \theta \cdot m_0}{t+\theta}, \frac{\sigma^2}{t+\theta}\right) \end{aligned}$$

and that the marginal likelihood satisfies

$$\begin{aligned} -\log \mathbf{P}\left(\{x_s : 1 \leq s \leq t\}\right) &= \frac{t}{2} \cdot \log\left(2 \cdot \pi \cdot \sigma^2\right) + \frac{1}{2} \cdot \log\left(\frac{t+\theta}{\theta}\right) \\ &\quad + \frac{t \cdot \theta}{t+\theta} \cdot \frac{\left(\bar{x}_t - m_0\right)^2}{2 \cdot \sigma^2} + \sum_{s=1}^{t} \frac{\left(x_s - \bar{x}_t\right)^2}{2 \cdot \sigma^2} \\ &= c_{t,\theta} + \frac{t \cdot \theta}{t+\theta} \cdot \frac{\left(\bar{x}_t - m_0\right)^2}{2 \cdot \sigma^2} + \frac{t-1}{2 \cdot \sigma^2} \cdot v_t, \end{aligned}$$

where $\bar{x}_t$ is the sample mean of $\{x_s : 1 \leq s \leq t\}$, v_t is the sample variance of $\{x_s : 1 \leq s \leq t\}$, and $c_{t,\theta}$ is an explicit constant which is independent of the data. In any case, if we can compute the filtering distribution and the marginal likelihood, then by comparison to these explicit formulas, we can recover the sample mean and sample variance of the observed data (which are well-posed quantities irrespective of the validity of our nominal 'model' for them).

Kalman Solution

For this model, the Kalman updates look rather simple, and can be computed as

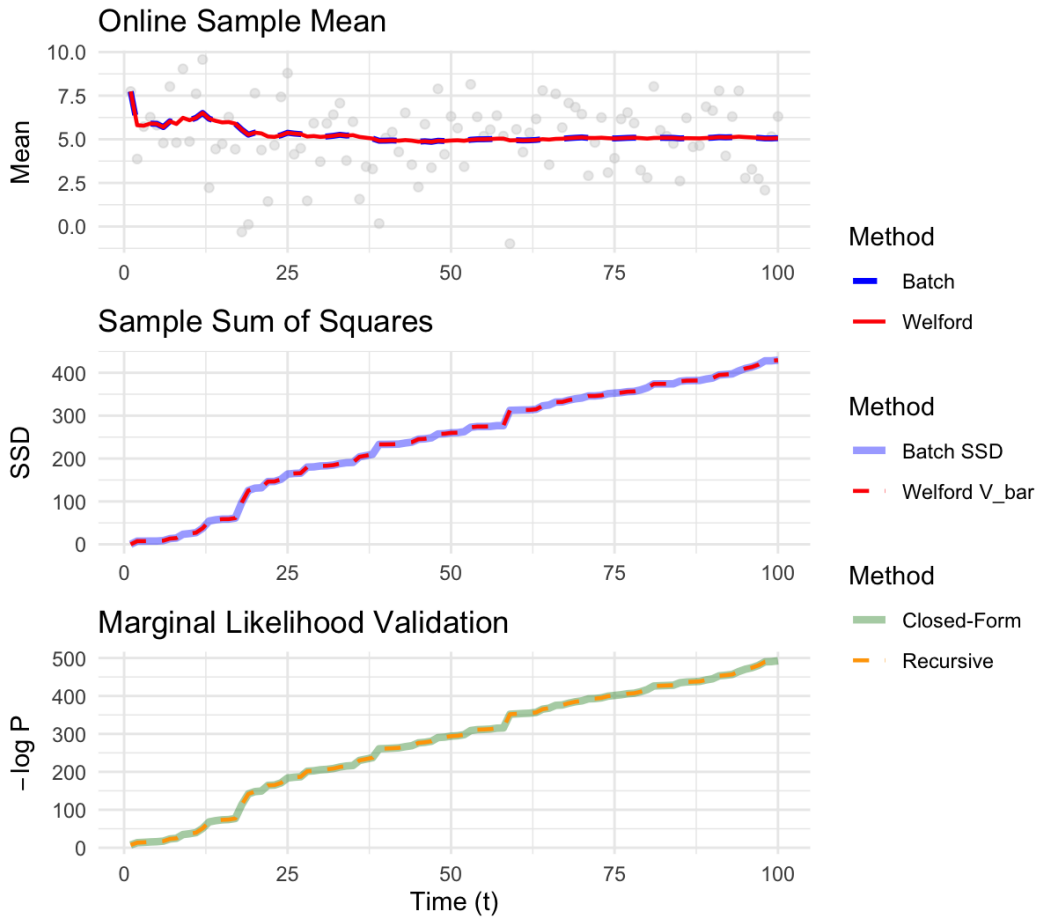
$$\begin{aligned} \text{for } t \geq 1, \qquad m_t &= m_{t-1} + \frac{1}{t+\theta} \cdot \left(x_t - m_{t-1}\right) \\ s_t^{-2} &= s_{t-1}^{-2} + \sigma^{-2} \\ V_t &= V_{t-1} + \frac{t+\theta-1}{t+\theta} \cdot \frac{\left(x_t - m_{t-1}\right)^2}{2 \cdot \sigma^2} \\ &\quad + \frac{1}{2} \cdot \log\left(2 \cdot \pi \cdot \sigma^2\right) + \frac{1}{2} \cdot \log\left(\frac{t+\theta}{t+\theta-1}\right). \end{aligned}$$

In particular, letting θ 'diverge[1] to 0', omitting the (data-independent) filtering variance updates, setting $2 \cdot \sigma^2 = 1$, and defining $\bar{V}_t = V_t - c_{t,\theta}$, we obtain the Welford-type recursions

$$\text{for } t \geq 1, \qquad m_t = m_{t-1} + \frac{1}{t} \cdot (x_t - m_{t-1})$$
$$\bar{V}_t = \bar{V}_{t-1} + \frac{t-1}{t} \cdot (x_t - m_{t-1})^2 .$$

Below, we present a simple numerical verification of the (well-known) fact that these recursions exactly reproduce the streaming means and variances.

Figure 1: A numerical verification of the online computations against the batch computations.



One benefit of the Kalman-type approach is that it allows for single-pass computations of these statistics in a streaming setting. Another is that the recursions are rather numerically stable; this is to be contrasted with the catastrophic cancellations which can plague more naive computations of these quantities. In the multivariate case, their stability requires more care, but can be addressed easily by drawing on insights from stabilising the Kalman filter (e.g. the square-root form).

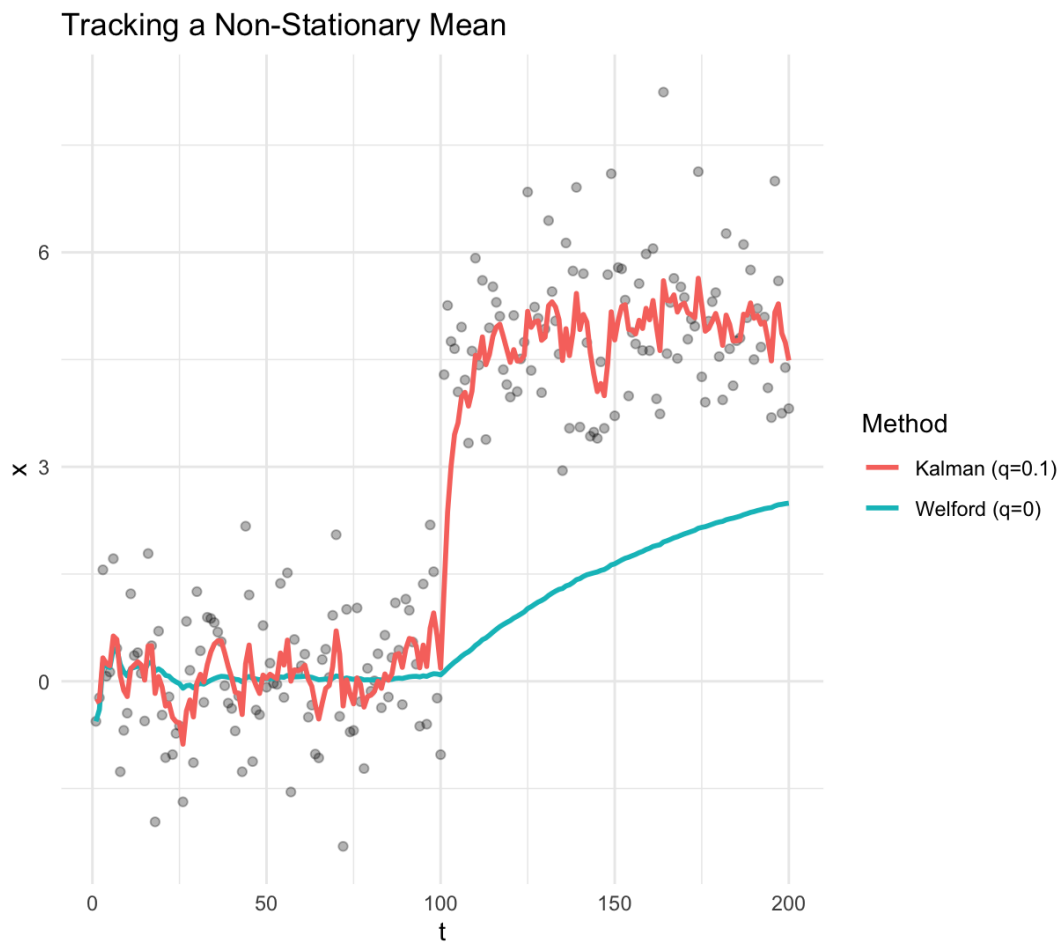
Some Extensions

A fun observation is that the Kalman filter can also be applied on trees, and so this would suggest a natural fork-merge approach to computing the mean and variance in a distributed manner. This is related to what is known variously as the Chan, Golub, and LeVeque method. One can envision this being useful for e.g. merging clusters in clustering algorithms.

[1] This can be thought of as letting the prior variance diverge to infinity, i.e. taking the limit towards an improper prior.

Another interesting variant is to account for non-stationarity by modelling the conditional mean of the observations as evolving slowly over time, following e.g. $\mu_t \sim \mathcal{N}(\mu_{t-1}, q)$ for some $q > 0$. The Kalman filter then computes an exponentially-weighted moving average of the observed $\{x_t : t \geq 1\}$, and the corresponding 'variance' parameter takes on a different interpretation, related to how well x_t can be predicted on the basis of $\{x_s : 1 \leq s < t\}$. Figure 2 demonstrates how the model-based perspective allows for adaptation to non-stationarity in data over time (though this is really a question of how to model, or what to compute, rather than how to compute an exact quantity efficiently and stably).

Figure 2: Tracking a non-stationary stream by incorporating process noise



Finally, some related algorithms exist for tracking of higher-order central moments of the data (e.g. Pébay's Algorithm), but the connection to the Kalman filter becomes somewhat weaker. It stands to reason that a method for tracking cumulants may also exist, but I have not been able to find this in the literature.